MILITARY SPECIFICATION
GYROSCOPE, RATE SWITCHING MS-17399

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification is used in aircraft to monitor vertical gyros during aircraft turns.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods
MIL-C-4150	Case, Carrying and Storage, Shock and Waterproof
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft, Electrical Equipment
MIL-L-6085	Lubricating Oil, Instrument, Aircraft Low Volatility

FSC 6610

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MIL-T-18303	Test Procedures; Preproduction and Acceptance For Aircraft Electronic Equipment, Format For
MIL-N-18307	Nomenclature and Nameplates For Aeronautical Electronic and Associated Equipment
MIL-A-22858	Attitude Indicating System, Remote, Standby, Two-Inch
MIL-G-23081	Gyro Indicator System, Vertical
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-B-81744	Barrier Film Solvents and Solutions

STANDARDS

Military

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirement For Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program For Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures For Packaging and Packing of
MIL-STD-838	Lubricating of Military Equipment
MIL-STD-1334	Barrier Film Coating for Anti-Friction Bearings

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MS-3106	Connectors-Electrical, Plugs, Straight
MS-17399	Gyroscope, Rate Switching

Federal

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements For
AR-34	Failure Classification For Reliability Testing, General Requirement For
AS-2844	Lubricating Oil, Instrument, Ball Bearing, High Flash Point

DOD Documents

SD-6	Provisions Governing Qualification (Qualification Products List)

*2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications, required by contractors in connection with specific procurement functions, may be obtained upon application to the Commanding Officer, Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

*3.1 Qualification - The instruments furnished under this specification shall be a product which has been tested and meets the quality assurance provisions specified herein and has been listed on or approved for listing on the applicable Qualified Products List in accordance with Provisions governing qualification SD-6.

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3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry, to be included therein, shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of the Rate Switching Gyroscope did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

* 3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

* 3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

3.3 Design and Construction - The Rate Switching Gyroscope shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the Rate Switching Gyroscope, excluding cables, shall not exceed 1.6 pounds.

* 3.3.2 Reliability - The contractor shall conduct a Reliability Program using MIL-STD-785 as a guide. On a reorder from a supplier, who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

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* 3.3.2.1 Operational Stability - The Rate Switching Gyroscope shall operate with satisfactory performance, continuously or intermittently, for a period of at least 2000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The Rate Switching Gyrsocope shall have a total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring schedule replacement shall be specified by the contractor.

* 3.3.2.3 Reliability In Mean-Time-Between-Failures (MTBF) - The equipment shall have 2000 hours of Mean (operating) Time Between Failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The Rate Switching Gyroscope shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The Rate Switching Gyroscope shall be capable of satisfactory operation, using external wiring, in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated, as necessary, to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors, attached to the cables, shall not be supplied as part of the equipment.

3.3.3.3 Electrical Connector - A hermetically sealed connector shall be affixed to the case. The connector shall mate with connector MS-3106-14S-5S. The pin connections shall conform to Table I.

TABLE I

Pin	Circuit
A	115V, 400 cps Single phase AC Power
B	115V, 400 cps Single phase AC Power
C	Turn Switching Relay Contact Terminal (N.C.)
D	Turn Switching Relay Contact Terminal (COM.)
E	Turn Switching Relay Contact Terminal (N.O.)

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* 3.3.4 Interchangeability - The equipment shall meet the interchangeability requirement of Specification MIL-E-5400.

3.3.5 Electromagnetic Interference - The instrument shall meet the interference requirements of MIL-STD-461.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and equipment identification marking shall be in accordance with MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 5°C)
Altitude	Normal Ground (0 to 5000 ft.)
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input Power	115 ± 1.0 VAC 400 ± 20 cps (AC Power Single Phase)

3.3.9 Service Conditions - The Rate Switching Gyroscope shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions, as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.9.1 Vibration - The Rate Switching Gyroscope shall operate satisfactorily when subjected to the vibration requirements of Curve IVA of MIL-E-5400.

3.3.10 Primary Input Power Requirements - The Rate Switching Gyroscope shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources, with characteristics as defined in MIL-STD-704, having limits as specified therein. The power required shall not exceed the specified amount:

(1) AC Power (Single Phase) 115V, Category "B", 10VA

3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the gyroscope.

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3.4 Performance - The Rate Switching Gyroscope shall provide switching during turns within specified time intervals at specified turn rates. Design shall be such as to provide unswitching within prescribed time intervals after return to zero turn rate from a particular rate of turn. The damping medium used in the Rate Switching Gyroscope shall be a viscous fluid (not to be construed as a gas). The Rate Switching Gyroscope shall be constructed to withstand, without failure, normal shocks, vibrations, and such other conditions as are incident to service, shipping, storage, and installation.

* 3.4.1 Performance Compatibility - The Rate Switching Gyroscope shall operate satisfactorily, in both the normal and the inverted positions, when installed in aircraft and shall meet the specified tests and requirements of this specification. The gyroscope shall also conform to all applicable requirements of MIL-A-22858 and MIL-G-23081. It shall be the supplier's responsibility to prove the components performance compatibility to the tests and requirements of this specification and to the applicable system requirements of MIL-A-22858 and MIL-G-23081 for all Qualification, Initial Production, Production Sampling Test Plans, and Reliability Assurance Tests.

3.5 Detail Requirements

* 3.5.1 Function - The Rate Switching Gyroscope shall provide switching during turns within specified time intervals at specified turn rates. Design shall be such as to provide no switching within specified time intervals after return to zero turn rate from a particular rate of turn. The Rate Switching Gyroscope shall consist of three assemblies:

(1) Gyroscope Assembly

(2) Hydraulic Viscous Damping Assembly

(3) Electronics Assembly

Each assembly shall be separately independent and concentric with the other. The Hydraulic Damping Medium shall conform with the requirements of 3.6.2. The Gyroscope Assembly shall conform with filling medium requirements of 3.6.1. With Hydraulic Viscous Damping, the mechanism shall always be free from hangup. Where electronic damping is used, the components, therof, shall comply with the specifications AR-5 and AR-10 and the performance requirements of this specification shall be met. The Rate Switching Gyroscope shall operate normally when properly connected in the gyroscope platform system, conforming to the requirements of this and other applicable specifications.

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3.5.2 Switching Mechanism - The switching mechanism shall contain a single pole double throw switch, with solid platinum alloy contact points or equivalent. The normally open circuit shall close and the normally closed circuit shall open when a turn rate of 15° per minute, or more, is imposed on the Gyroscope. The contact rating shall not be less than 2 amperes resistive load at 115 volts. The contact circuits shall be electrically isolated from the other circuitry in the gyroscope and from the case.

3.6 Case - The case shall be hermetically sealed and shall be constructed so that the mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures. The hermetic seal shall be perfectly closed and an airtight seal made between metallic and metallic, or metallic and vitric materials. The hermetic seal is not intended to include seals accomplished by gaskets. The dimensions of the case shall conform to MS-17399.

3.6.1 Filling Medium - The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be a mixture of 95 percent helium and 5 percent carbon dioxide, or a filling medium approved by the procuring activity. The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.6.2 Hydraulic Medium - The Hydraulic Filling Medium shall be Dow Corning 510 fluid or equivalent and shall be filtered so that the maximum particle size shall not exceed 8.8 Microns diameter in the medium.

3.7 Degree of Cleanliness - The Rate Switching Gyroscope shall be assembled in a Class 100 clean room, as specified in FED-STD-209. The unit shall be thoroughly cleaned to assure that close tolerance interface moving surfaces will not malfunction because of contaminating deposits or loose particles resulting from environmental operating conditions.

* 3.8 Lubrication - The Gimbal bearings shall be lubricated with oil AS-2844 in accordance with MIL-L-6085 unless immersed in the hydraulic filling medium of paragraph 3.6.2. The spin rotor bearing synthetic separators shall be impregnated with AS-2844 lubricating oil. The Gyroscope Spin Rotor Bearings shall otherwise be lubricated with grease MIL-G-81322 or with Minipure grease. Particle size not exceeding 25 microns for 2% volume shall be used in accordance with MIL-STD-838. All ball bearings shall be ABEC-7 grade or greater.

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* 3.9 Barrier Film Coating - All anti-friction bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with Specification MIL-B-81744. Bearing chamfers and corners shall also be barrier coated to prevent bridging and migration of the lubricant from the bearing working area.

* 3.10 Warm-Up Time - The Gyroscope shall not require a warm-up period of more than 90 seconds under any temperature conditions specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Qualification Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Test

* 4.2 Qualification Tests - The qualification inspection shall consist of all the tests of 4.7.

* 4.2.1 Qualification Samples - Qualification Inspection Samples shall consist of three samples. The samples, submitted, shall have been previously subjected only to the individual tests. These samples shall be forwarded, at the applicant's expense to the laboratory designated in the letter of authorization (Section 6.6).

* 4.2.1.1 Qualification Inspection Sample Identification - The Qualification Inspection Samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

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Samples for Qualification Inspection (Name of Equipment) (Submitted by [Manufacturer's name, date, and part number] for Qualification Inspection in accordance with Specification MIL-G-23083 under authorization [reference letter authorizing tests]).

4.2.2 Production Equipments - Equipments supplied under the contract shall, in all respects including design, construction, workmanship, performance and quality, be equivalent to the approved qualification sample. Each equipment shall be capable of successfully passing the same tests as imposed on the qualification sample. Evidence of noncompliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production Rate Switching Gyroscopes shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This gyroscope shall be selected by the procuring activity after the gyroscope has successfully passed all individual tests.

* 4.3.1 Scope of Tests - The Rate Switching Gyroscope, selected for Initial Production Tests, shall be subjected to all the tests of 4.7.

4.3.2 Accessory Material - In addition to the complete Rate Switching Gyroscope, submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the gyroscope.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of Rate Switching Gyroscopes under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on gyroscopes previously accepted when requested by the procuring activity.

4.3.4 Initial Production Sample Identification - The Initial Production Sample shall be plainly identified by durable tags, securely attached and marked for the following information:

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Sample for initial production Testing Gyroscope, Rate Switching (Type number if any) Submitted by (Manufacturer's name, date) for initial production testing in accordance with Specification MIL-G-23083 under Contract or Order number _____.
MS17399 - _____ (add proper dash number)
(Manufacturer's name, Type, Part number)

4.3.5 Reconditioning of Initial Production Test Sample - On completion of the Initial Production Test, the gyroscope shall be reworked by the contractor by replacing all wear and damaged items. After reworking, the contractor shall resubmit the gyroscope for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors, not having testing facilities satisfactory to the procuring activity, shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative detail results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

* 4.4.1 Individual Tests - Each Rate Switching Gyroscope, submitted for acceptance, shall be subjected to the Individual Tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum requirement, each gyroscope accepted shall have passed the following tests:

(1) Examination of Product	4.7.1
(2) Power Consumption	4.7.2
(3) Case Leakage	4.7.3
(4) Switching Rate	4.7.4

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(5) Yaw Oscillation	4.7.5
(6) Steady State Turn and Yaw Oscillation	4.7.6
(7) Power Failure	4.7.7

4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A Tests. The test sample selected for sampling tests by a Government inspector shall have first passed the individual tests. The test samples which have been subjected to sampling tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the individual tests. Selection of units shall be on the following basis:

Quantity of Gyroscopes Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
	1 for each additional 100 or fraction thereof

When an instrument failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each 15 from those still on hand, and proceed with sampling as indicated above when the next quantity is offered for acceptance. One additional sample, selected at random, shall be tested from each monthly lot for shelf life as per paragraph 4.11. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. Concurrent delivery shall be permitted during the storage period.

* 4.4.2.1 Scope of Tests - As a minimum, each Rate Switching Gyroscope, selected for sampling tests, shall be subjected to the following tests:

(1) Individual Tests	
(2) Insulation Resistance	4.7.8
(3) Electronic Interference	4.7.9
(4) Voltage and Frequency Variation	4.7.10

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(5) Voltage Transient Surge	4.7.11
(6) High Altitude Low Temperature	4.7.12
(7) Performance Compatibility	4.7.13
(8) Low Temperature Operation	4.7.14
(9) High Temperature Operation	4.7.15
(10) Vibration	4.7.16

* 4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by the Production Acceptance Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and AR-34.

* 4.4.3.1 Reliability Production Acceptance Phase Tests - Throughout production, the equipment shall be tested as outlined in MIL-STD-781 under the section entitled "Production Acceptance Phase of Production Reliability Tests."

* 4.4.3.2 Reliability Acceptance Phase - Reliability production acceptance testing shall be conducted through an acceptance burn-in. Each equipment produced but not submitted to the qualification test shall be operated for 120 operating hours burn-in, and the test data (operating hours and failures) shall be compared to the accept/reject criteria of Test Plan II of MIL-STD-781. Delivery of equipment shall not be held up to await a decision under the test plan. In event an accept decision is reached, the test duration of each equipment shall be reduced to 80 operating hours. At any time that the current total of test hours and test failure plotted in Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop acceptance of the equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation shall show other than a rejection decision. For the production acceptance phase, test level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5g with equipment hard mounted.

* 4.4.3.3 Test Details - The test details such as the length of the heat portion of the temperature cycle, special failure criteria, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The minimum performance characteristics to be measured shall be as specified in 4.4.3.4. The duty cycle shall be as specified in 4.5.1.1.

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* 4.4.3.4 Performance Characteristic to be Measured - The following performance characteristic will be measured and recorded, at least daily, during the high temperature stabilization period:

Switching Rate (4.7.4)
Yaw Oscillation (4.7.5)
Steady State Turn and Yaw Oscillation (4.7.6)
Power Failure (4.7.7)

If the value of any specified equipment parameter is not within tolerance, a failure shall be recorded and required action shall be taken. The failure shall be presumed to have occurred immediately after the last successful measurement of the same parameter.

4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (Paragraph 4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests, no preventive maintenance measures may be performed upon the Rate Switching Gyroscope.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of Rate Switching Gyroscopes for the purpose of checking the effect of any design or material change on the performance of the gyroscopes and to assure adequate quality control. The gyroscopes selected for special tests may be selected from gyroscopes previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of gyroscopes for special tests shall be made as follows:

(1) On a gyroscope manufactured after an engineering or material change. (Within a block of 10 units).

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

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4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the production tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the Individual Test to check all gyroscopes for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 1500 hours duration and shall be conducted on Rate Switching Gyroscopes that have passed the Individual Tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following. (Gyroscopes which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.):

Quantity of Gyroscopes Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
AC Voltage	115 ± 5 Volts (at lowest applicable frequency 380 Hz)
Mounting	Normal and Inverted Positions

* 4.5.1.1 Operating Test Conditions - The gyroscope shall be mounted on a rate-of-turn table that is capable of automatic periodic reversing with a pause at each change in the direction of rotation. The table shall turn at a rate sufficient to cause the gyroscope to switch, causing the internal circuit between pins D and E to close and the circuit between pins C and D to open. The pause at each change in direction of rotation shall be sufficient in duration so that the gyroscope will unswitch causing the internal circuit between pins D and E to open and the circuit between pins C and D to close. The test current load, at pin D of the rate switching gyroscope, shall be 25 MA or equivalent to the displacement gyroscope roll erection circuit current of MIL-G-23081 or the vertical gyroscope of MIL-A-22858, whichever is greater. The mounting positions (normal or inverted) shall be reversed every 24 hours.

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the rate switching gyroscope temperature to stabilize. Once each day, the gyroscope shall be turned on and off several times and subject to the performance check of 4.5.3.

* 4.5.3 Performance Check - Once each day, during the test, the gyroscope shall be subjected to and meet the requirements of 4.7.2 (Power Consumption), 4.7.3 (Case Leakage), 4.7.4 (Switching Rates), 4.7.5 (Yaw Oscillation), 4.7.6 (Steady State Turn and Yaw Oscillation), and 4.7.7 (Power Failure).

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the Rate Switching Gyroscope, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number.

(2) The circuit reference symbol number.

(3) The part function.

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(4) Name of manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design change or material corrections for all failures which have occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - A Rate Switching Gyroscope, which has been subjected to the life test, shall be reconditioned as follows:

(1) On completion of the life test, the gyroscope shall be reworked by the contractor by replacing all "wear items." The "wear items" shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the gyroscope for acceptance.

4.6 Test Procedures - The procedures used for conducting Initial Production, Acceptance, and Life Tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

* 4.7 Test Methods - Each time the test of Paragraphs 4.7.2, 4.7.4, 4.7.5, 4.7.6, or 4.7.7 are required, the gyroscope shall be tested in the normal and inverted mounting position.

4.7.1 Examination of Product - Each Rate Switching Gyroscope shall be examined carefully to determine that the material and workmanship requirements have been met.

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* 4.7.2 Power Consumption - The power consumption of the Rate Switching Gyroscope will be measured and shall not exceed 10VA, starting and running.

* 4.7.3 Case Leakage - The Rate Switching Gyroscope shall be tested for case leakage with a mass spectrometer or an equivalent, quantitative leak test. The rate of leakage of the gas or mixture of gases in the case, adjusted to a pressure differential of one atmosphere, shall not exceed one micron cubic foot per hour.

* 4.7.4 Switching Rate - The Rate Switching Gyroscope, with power applied and operating for not less than one minute, shall be mounted on a rate table and caused to turn at a rate of 15 degrees per minute. At this rate of turn the internal circuit between pins D and E shall close and the circuit between pins C and D shall open within 30 seconds. This procedure shall be repeated at 20 degrees per minute, 45 degrees per minute, and 180 degrees per minute rates of turn. The same results shall occur within 20 seconds, 8 seconds, and 4 seconds, respectively. The switching rates shall be within the same limits regardless of the direction of turn. Each consecutive switching rate measurement must be run in alternating clockwise and counterclockwise directions. The contact circuits shall return to normal unswitch within 15 seconds after the turn rate has ceased.

4.7.5 Yaw Oscillation - The Rate Switching Gyroscope, with power applied and operating for not less than one minute, shall be mounted on a yaw test table. At zero rate turn, the circuit between pins D and E shall remain open and the circuit between pins C and D shall remain closed when yaw oscillations, of ± 1.0 degree with a two-second time period, are applied.

* 4.7.6 Steady State Turn and Yaw Oscillation - The Rate Switching Gyroscope, with power applied and operating for not less than one minute, shall be mounted on a test apparatus capable of superimposing a ± 1.0 degree yaw oscillation with a two-second time period on a 30 degrees per minute rate of turn. First the yaw oscillation shall be applied and the unit shall remain unswitched (in effect, the circuit between pins D and E shall remain open and the circuit between pins C and D shall remain closed). With the yaw oscillation still present, the 30 degrees per minute rate of turn shall be applied and within 30 seconds the unit shall switch (in effect, the circuit between pins D and E shall close and the circuit between pins C and D shall open). After the unit has indicated a switch condition, wait 30 seconds to one minute before removing the steady state turn. The unit shall remain switched until the steady state turn is removed. After the steady state turn has been removed, and with the yaw oscillation still present, the unit shall unswitch and remain unswitched within 15 seconds. These results shall be the same regardless of the direction of turn.

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4.7.7 Power Failure - The Rate Switching Gyroscope, with power applied and operating for not less than one minute, shall be mounted on a rate table and caused to turn at a rate sufficient to cause switching. After switching has occurred, power to the gyroscope shall be removed. The circuit between pins D and E shall open and the circuit between pins C and D shall close immediately after removal of power.

4.7.8 Insulation Resistance - A potential of 500 VDC shall be applied between isolated pins and between pins and the case for a period of five seconds. There shall be no breakdown of insulation. The resistance between circuits and chassis shall not be less than 50 megohms.

* 4.7.9 Electromagnetic Interference - The Rate Switching Gyroscope, properly connected and operating, shall be tested in accordance with MIL-STD-461, Class ID equipment, methods CE01, CE02, CE03, CE04, CS01, CS06 (limited to maximum frequency of 400 MHz), RS01 and RS03. At the conclusion of this test, the gyroscope shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.10 Voltage and Frequency Variation - The gyroscope shall be subjected to the variations of voltage and frequency of 4.7.10.1 and 4.7.10.2 and shall meet the requirements of 4.7.2 (Power Consumption), 4.7.4 (Switching Rate), 4.7.5 (Yaw Oscillation), 4.7.6 (Steady State Turn and Yaw Oscillation), and 4.7.7 (Power Failure).

4.7.10.1	AC Volts	Frequency Hz
	108	380 and 420
	118	380 and 420

4.7.10.2	AC Volts	Frequency Hz
	104	360 and 440
	122	360 and 440

4.7.10.3 The gyroscope shall be run for 10 minutes at each condition of the voltage and frequency as shown below:

AC Volts	Frequency Hz
102	370 and 430
124	370 and 430

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During these conditions, the gyroscope:

(1) May have momentary loss of function; however, this momentary loss shall not affect the gyroscope performance later.

(2) After the gyroscope returns to normal operation, it shall recover automatically to specified performance.

At the conclusion of this test, the gyroscope shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.11 Voltage Transient Surge - The gyroscope shall be subjected to the 180 VAC and the corresponding time, of Curve 1 of Figure 3, Page 18, of MIL-STD-704. At the conclusion of this test, the gyroscope shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.12 Measurement and Weight - The gyroscope shall be weight and case dimensions measured. The results shall be recorded.

(1) The case dimensions shall conform to MS17399.

(2) The weight shall not exceed 1.6 pounds.

* 4.7.13 Performance Compatibility - The Rate Switching Gyroscope, properly connected with other system components of MIL-A-22858(AS), shall perform its designed function without causing degradation in the performance of the other system components. The Vertical Gyroscope shall be precessed off vertical 20 to 30 degrees in roll (right bank position). The position and movement of the vertical gyroscope shall be monitored on the attitude indicator. The Rate Switching Gyroscope shall be exposed to a rate of turn which will cause it to switch. After switching has occurred, the attitude indicator shall be observed for evidence that the vertical gyroscope has ceased erecting. The rate of turn shall then be removed and the attitude indicator observed for evidence that erection has resumed. The same procedure shall be repeated with the system components of MIL-G-23081.

* 4.7.14 Low Temperature Operation - The gyroscope shall be properly connected and tested in accordance with Step 2, Temperature and Altitude Test of MIL-T-5422, for a period of four hours. At the end of the four-hour period and while the gyroscope is still at the low temperature, power shall be applied. After power has been applied for 90 seconds, the gyroscope shall meet the requirements of Paragraphs 4.7.2 (Power Consumption, 4.7.4 (Switching

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Rates), 4.7.5 (Yaw Oscillation), 4.7.6 (Yaw Oscillation and Steady State Turn), and 4.7.7 (Power Failure). The "Time To Switch" for the turn rates, specified in 4.7.4, shall not exceed 65, 45, 20, and 10 seconds, respectively. The switching contact shall return to normal within 80 seconds while the gyroscope is at -54°C ± 2°C. At the conclusion of this test, the unit shall be returned to room temperature and shall meet the requirements of the Individual Tests.

* 4.7.15 High Temperature Operation - The gyroscope shall be properly connected and tested in accordance with Steps 6 and 7, Temperature and Altitude Tests, of MIL-T-5422. For both steps, the gyroscope shall meet the requirements of Paragraphs 4.7.2 (Power Consumption), 4.7.4 (Switching Rates), 4.7.5 (Yaw Oscillation), 4.7.6 (Steady State Turn and Yaw Oscillation), and 4.7.7 (Power Failure). After the gyroscope has returned to room temperature, it shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.16 Vibration - The Rate Switching Gyroscope shall be tested in accordance with Part I, Vibration Tests, of MIL-T-5422, except the frequency range shall be from 5 to 2000 Hz. The vibration test equipments employed, shall not apply a yaw rate greater than 30 degrees per minute to the rate gyro. There shall be no switching, damage or irregular operation of the gyroscope during the Vibration Test. At the conclusion of this test, the gyroscope shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.7.17 High Temperature Exposure - The High Temperature Exposure Test shall be conducted in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422, except the time shall be for a period of 24 hours, with the gyroscope properly connected but with no power applied. At the end of this period and while still at the stated high temperature, the gyroscope shall meet the requirements of Paragraphs 4.7.2 (Power Consumption), 4.7.4 (Switching Rate), 4.7.5 (Yaw Oscillation), 4.7.6 (Steady State Turn and Yaw Oscillation), and 4.7.7 (Power Failure). There shall be no sticking or erratic operation. After the gyroscope has returned to room temperature, it shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.18 High Altitude - Low Temperature - The Rate Switching Gyroscope, properly connected but with no power applied, shall be subjected to the following steps of the Temperature and Altitude Tests of MIL-T-5422, except as modified below: Step 1 for 72 hours and Step 2 for 72 hours. After exposure to -54°C (-65°F) for 72 hours, the pressure in the chamber shall be reduced to the equivalent of 100,000 ± 500 feet and power shall be applied to the gyroscope. The gyroscope shall meet the requirements of Paragraphs 4.7.2 (Power Consumption) and 4.7.4 (Switching Rate). The "time to switch" for the turn rates, specified in 4.7.4, shall not exceed 65, 45, 20, and 10 seconds, respectively. The switching contacts shall return

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to the normal position within 80 seconds when the turn rates are ceased at this temperature and pressure. After the gyroscope has returned to room temperature and pressure, it shall be subjected to and meet the requirements of the Individual Tests.

* 4.7.19 Temperature Shock - The Rate Switching Gyroscope shall be tested in accordance with Procedure I of MIL-E-5272, except the temperature range shall be -54°C to 95 °C. After the gyroscope has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The gyroscope shall be subjected to and meet the requirements of the Individual Tests.

4.7.20 Impact Shock - The Rate Switching Gyroscope shall be tested in accordance with the Shock Test of MIL-T-5422. At the end of this test, the gyroscope shall be subjected to and meet the requirements of the Individual Tests.

4.7.21 Acceleration - The Rate Switching Gyroscope, not operating, shall be mounted on a centrifuge in its normal manner and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The gyroscope shall be subjected to an acceleration of 20 g's in each of the above axes for a period of one minute. At the end of this acceleration, the gyroscope shall be subjected to and meet the requirements of the Individual Tests. No damage to the gyroscope shall result from this test.

4.7.22 Salt Spray - The Rate Switching Gyroscope shall be tested in accordance with the Salt Spray Test of MIL-T-5422, except that the test period shall be 50 hours. At the end of the 50-hour period, the gyroscope shall be subjected to and meet the requirements of the Individual Tests.

4.7.23 Fungus Resistance - The Rate Switching Gyroscope Assembly shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422, except that the test period shall be 14 days. After the testing procedure has been completed, any evidence of the growth of fungi found upon inspection shall be cause of rejection of the gyroscope assembly or of lots represented by the test gyroscope assembly. After completion of the test procedure, the gyroscope assembly shall be subjected to and meet the requirements of the Individual Tests.

4.7.24 Humidity - The Rate Switching Gyroscope Assembly shall be tested in accordance with the Humidity Test of MIL-T-5422, except 71°C shall be used in lieu of 50°C in Steps 1 and 2. At completion of the test procedure, the gyroscope assembly shall be subjected to and meet the requirements of the Individual Tests.

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4.7.25 Internal Examination - The case of the instrument shall be opened and the internal mechanism examined for compliance with the contract, detail specification, and all applicable specifications. Any deterioration or damage which could in any manner prevent the instrument from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.7.26 Shelf Life - The instrument shall be placed in storage for a period of not less than 30 days. During the 30-day period, the instrument shall remain dormant as practical. The temperature shall remain constant, however, normal room temperature variations will be permitted. At the end of the 30-day period, the instrument shall be subjected to and meet the Individual Test of paragraph 4.4.1.

4.8 Reconditioning of Tested Gyroscopes - The Rate Switching Gyroscopes, which have been subjected to Acceptance and Life Tests, shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the gyroscopes for acceptance.

4.9 Presubmission Testing - No item, part, or complete gyroscope shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable specification requirements.

4.10 Rejection and Retest - The Rate Switching Gyroscopes, which have been rejected, may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full details concerning previous rejection and the action taken to correct the defects found in the original rejected gyroscope shall be furnished to the Government Inspector.

4.11 Associated Equipment - The Rate Switching Gyroscope manufactured in accordance with this specification shall be compatible with Vertical Gyroscopes conforming with Specifications MIL-A-22858 and MIL-G-23081.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Each Rate Switching Gyroscope shall be packaged in accordance with MIL-STD-794, Level A or C, as specified in the contract or order (See 6.7). For level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method 1C, utilizing a rectangular container. The container shall conform to MIL-C-4150.

5.2 Packing - Each Rate Switching Gyroscope shall be packed in accordance with MIL-STD-794, Levels A, B, or C, as specified in the contract or order (see 6.7). The package and exterior shipping container may be one and the same container.

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5.3 Rough Handling Tests - Rough Handling Tests shall apply to all levels of packaging and packing. The tests performed on a prototype example shall be Rough Handling, Dummy Load and Production Tests of MIL-STD-794. No vibration tests are required. Two pilot shipping containers are required to be tested. The heights of drop shall be 30 inches from the free-fall flat drop and 36 inches for the free fall cornerwise drop. All testing shall be done on all axes, in each direction, in each axis, and with actual equipment, as packaged for shipment. After tests, the equipment shall not suffer damage or subsequently fail to provide performance as specified. All testing shall be done at the Contractor's plant. The Contractor shall notify the Naval Air Systems Command, in sufficient time, in advance of prototype testing to permit government representation at the test site.

5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129 and as follows:

5.4.1 Precautionary Marking - The following precautionary markings, warning of the fragility of the item, shall appear on two opposite sides of each interior package and shipping container. Containers shall be marked with arrows on two opposite sides of each interior package and shipping container to indicate which surface must be always up:

FRAGILE
DELECATE INSTRUMENTS
HANDLE WITH CARE
UP

In addition, containers shall be marked "Reusable, OPEN WITH CARE" in letters at least two inches high and on two opposite sides of each container.

6. NOTES

6.1 Intended Use - The Switching Rate Gyroscope, covered by this specification, is a component of the Vertical Gyroscope Indicator System (MIL-G-23081) and is intended for use in high performance aircraft to sense turns above a specified rate and switch erection circuits of the displacement gyroscope during these turns. This Switching Rate Gyroscope is used in the same manner, also, as a component of the Remote Attitude Indicating System (MIL-A-22858).

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6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ± 1.0 VAC 400 Hz. These data are to be used in testing the Rate Switching Gyroscope at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this Rate Switching Gyroscope. Where it appears, a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and rates should be so selected that the life of the subassembly is at least four times greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5000 hours and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedences shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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* 6.6 Qualification - With respect to products requiring qualification, awards will be made only for each product as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. Suppliers are directed to the provisions governing qualification (SD-6) and are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is The Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360 and information pertaining to qualification of products may be obtained from that activity.

* 6.7 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

(3) Monthly reliability reports in accordance with MIL-STD-781.

6.8 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

Project Number 6610-N045

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G-23083C(AS) GYROSCOPE, RATE SWITCHING MS-17399

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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